SAQ - Paper-I

(Group-A)

Biophysics

Laws & Equation

1. **State & deduce with explanation & importance-**
   a) Write the H-H equation for carbonic acid buffer solution **(CMU: May-25, 23,** Nov-20), calculate the pH of this solution of half neutrilization **(CMU:**Nov-19)
   b) Law of mass action. or Give a mathematical deduction of dissociation of carbonic acid and its use in clinical chemistry (Ju-13, Ja-09,11,15)
   c) H-H equation (May-17, Nov-17, Ja-06, 12,13, 14, 16) **(CMU:** May-20, Nov-19) with it's clinical importance. **(CMU:** Nov-21, 20) (Nov-18) How does it explain acid base status of human body? **(CMU:** May-23, Nov-23) (Nov:16, Ja-13)
   d) Gibbs-Donnan equilibrium. (Ja-09, 07, Ju-09, 10, 11)
2. **How can you get/ derive-**
   a) The definition of acidosis & alkalosis from H-H equation. **(CMU:** Nov-23, May-20) (Ja-09,Ju-10,11)
   b) Idea of pH from laws of mass action.
   c) H-H equation from law of mass action. (Ja-11)
   d) Formulate H_2CO_3 acid dissociation into an equation & relates pH with HCO_3 and PCO_2 (Ju-14)
3. Write the H-H equation with it's application in acid base balance **(CMU:** May-19) (Ju-16, May-17)

Buffer, pH, acid-base balance

1. Define acidosis & alkalosis. What do you mean by respiratory acidosis. (Ja-17)
2. What is anion gap? Write down the causes, compensation and correction of high aniongap metabolic acidosis. (Ja-17, May-18)
   (write the H.H Equation with its application in acid base balance) July-17
3. **Give the mode of action with diagram-**
   a) Hb buffer.(Ju-06, 09,12, Ju-15, May-17, Nov-17, 18) Classification of blood buffer?
   b) Plasma protein buffer. (Ju-09, 06,12)
   c) HCO_3^- buffer **(CMU: May-25,** Nov-22) (Ju-10,11, Ja-14)
   d) Mechanism of action of PO_4^{2-} buffer,NH_3 buffer in kidney.
   (Ja-17, May-18) **(CMU:** May-22). **(Nov-25)**
4. **Compare & contrast between:**
   a) Mode of action of HCO_3^- & phosphate buffer. (Ja-07)
   b) HCO_3^- & NH_3 buffer. (Ju-08) c) PO_4^{3-} & NH_3 buffer. (Ja-11, 12)
   d) Metabolic & respiratory acidosis (Ju-16, Ja-14)
5. Show the ABG change in metabolic acidosis. What are three common causes of metabolic acidosis? **(CMU:** Nov-22) **(May-25)**
6. **Give the biomedical picture of-**
   a) Compensated & uncompensated respiratory acidosis & alkalosis.
   b) Compensated & uncompensated metabolic acidosis (Ju-13) & alkalosis & define metabolic acidosis. **(CMU:** Nov-22).
   c) Show the changes in acid base parameters supporting metabolic acidosis and respiratory alkalocis **(CMU:** Nov-19)
7. Give the biochemical picture of respiratory acidosis and metabolic Alkalosis. **(CMU:** Nov-20)
8. **Mathematical problems:**
   i) Define pH. **(Nov-25)** Find out the pH of solution when concentration of $NaHCO_3$ are 0.025 M and H_2CO_3 0.00125 M respective (log 20=1.3) **(CMU:** May-19)
   ii) If the pH of a blood is 7.4 what will be the $[H^+]$ (Log 0.6=3.98)
   iii) Calculate the pH of a solution having $[H^+]$=0.00456 mmol/L(Given, log 4.56=0.6589) **(CMU:** May-21,18)

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iv) Define pH with its importance. **(Nov-25)** Calculate the [OH] in a solution with a (H^+] of 1.3×10^{-4}mol/L **(CMU:** May-22)

9. Write the composition of body buffer, plasma buffer & blood buffer. Classify buffer with example blood buffer (Nov-17). example of kidney buffer **(CMU:** May-22, 19) Give the mechanism of buffer action. **(CMU:** Nov-19) (Ju-14, May-19)
10. If pH of HCO_3 buffer solution is 6.1 then state the condition of buffer system (Nov-15)
11. Mechanism of action of buffer in human body: **(CMU:** May-19)
12. Classify blood buffer **(CMU:** Nov-19). How do they control acid-base balance. (Ja-16)
13. An elderly patient attend hospital with the following blood picture:
    i) pH= 7.20 ii) HCO_3= 25mmol/L iii) PCO_2 = 68 mmHg
14. Describe the acid base status of the patient.
15. Find out the pH of a solution when concentration of $NaHCO_3$ are 0.025 M and H_2CO_3 0.00125 M respectively (log20=1.3)
16. Define & classify buffer **(CMU:** May-21, 20) (Ju-16, Ja-17)
17. Define buffer Write in short the mechanism of buffer action. What is buffering capacity? (May-16, Nov-16)
18. Define pH. How normal blood pH is maintained? (May-16)
19. What is chloride shift mechanism **(CMU:** May-19) (May-17)
20. Name the plasma colloids and crystalloids. State the functions of crystalloids in human body **(CMU:** May-19)
21. Define buffer, name respiratory buffer & component of ABG Analysis. **(CMU:** Nov-22)
22. Why bicarbonate buffer is the most powerful buffer of blood? **(CMU:** May-20)
23. **SN:** a) Buffer. (Ja-07,14, Nov-15)
    b) Compensated respiratory acidosis.(Ju-09, Ja-08)
    c) pH (Ja-14)
    d) Compensated metabolic acidosis. (Ja-11)
    e) Metabolic alkalosis
    f) K_w
    g) Anion gap (Ja-13,14,16 Ju-10,11, 13,14,15,16, May-15,19, Nov-16)
    h) pK (Ju-15, May-19)
    i) Km (Ja-16)
    j) Weak acid (May-19)
    k) Buffering capacity **(CMU:** May-22)
    l) Respiratory alkaosis **(CMU:** May-22)
24. Name of buffers. Show with diagram the mechanism of action of phosphate buffer. **(CMU:** May-21)
25. What is the mechanism of action of bicarbonate buffer. **(CMU:** May-23, Nov-21) Why it is a powerful buffer of blood. **(CMU:** Nov-22)
26. Classify acid-base disorder with clinical example **(CMU:** May-23)
27. Define buffer name the blood buffers. Show with diagram the mechanism of action of hemoglobin buffer. **(CMU:** May-24)
28. Enumerate the simple Acid-base disorder with their primary defects and compensatory biochemical changes! Mention two important cause of metabolic acidosis. **(CMU: May-25)**
29. Determin the pH of water. **(Nov-25)**
30. Discuss the role of respiratory system in compensating metabolic acidosis. **(Nov-25)**

Solution

1. Define colloids? (May-16) with eg (May-19). & biomedical importance.
2. Compare & contrast between crystalloids & colloids. **(CMU:** May-22) (May-16, 19, Ja-09, 12, 13,17, Ju-10, 13,15,16)
3. Properties of colloid. **(CMU: Nov-25)** (Ja-13,16 Ju-13, Nov-16)

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4. Function of colloid in blood.(Ja-,09,10,12, Ju-13)
5. Give the difference between emulsoids & suspensoids. (Ja-07)
6. Define & classify solution. Difference between normal saline & normal solution. (Ju-11)
7. Define mole, equivalent, osmole.
8. Calculate the amount of each salt required to produce 100 ml 0.5 normal solution.
9. What is protective colloid. **(CMU:** May-22) → PDF-86
10. **Give the osmolarity of the following solution:**
    i) 0.9% normal saline. (Ju-10)
    ii) 0.5% molar urea. (Ju-10)
    iii) 5% glucose solution. (Ja-16)
11. Prepare 100ml 0.5% solution of glucose and calculate its osmolarity (May-16)
12. Give the calculation to prepare 100ml deci normal (N/10) HCl with Sp. gravity of 1.36 & purity 38%.(Ja-05, May-15, Nov-17)
13. Calculate the osmolarity of normal saline (MW of NaCl known as 58.5) (May-19) why it is called isotonic? (Ju-13,14)
    Or, mathmaticlly prove that normal saline is an isotonic solution. **(CMU:** Nov-24**)**
14. Define osmole (May-16) Mention the normal plasma osmolarity. How will you calculate it? (Ja-14)
15. How normal blood P^H is maintained (Ja-14)
16. Define colloid with example. Mention the important properties of colloids with significance and clinical important if any. (Ju-14)
17. Calculate how much gram of $CaCl_2$ salt will be required to produce 100 ml of .5 molar solution (molecular weight of $CaCl_2$=111) (Nov-15, Ju-16)
18. Write the clinical application of dialysis **(CMU:** May-24**)** (Ja-16)
19. Write down the important differences between Osmosis and Diffusion? **(CMU:** Nov-22) (May-17)
20. Prepare 10ml 5% urea solution calculates osmolarity with comment and examples. **(CMU-**May-19) (May-17)
21. Give the calculation for preparing 1000ml 5% glucose solution. Find its osmolarity. **(CMU:** Nov-20)
22. Prepare 1000ml 5% Dextrose solution calculate its Osmolarity with comment **(CMU:**Nov-19) (May-18)
23. Prepare 10ml 0.1M urea solution.
24. Give five differentiating points between colloids & crystalloids. Give example of each. **(CMU:** May-24, 20)
25. Define standard solution. Calculate osmolarity of normal saline (Molecular weight of NaCl=58.5gm) **(CMU:** May-20)
26. **SN:** a) Molar solution. (Ju-06)
    b) Plasma osmolarity **(CMU:** May-19).
    c) Isotonic Solution **(CMU:** Nov-22, 21, 20)
    d) Indicator. (Ja-07) e) Normal saline.(Ja-13, 14) f) Isoelectric pH. (Ja-11,13)
    g) Dialysis. **(CMU:** Nov-21, 19) (Ju-16, Ja-11,13,17)
    h) Thyxotrophy. (Ja-11) i) Standard solution. (Ja-12, Ju-14)
    j) Serum sodium (Ja-15)
27. Define crystalloid with example. What are the function of crystalloid. **(CMU:** Nov-21)
28. What is dialysis? **(CMU:** May-24**)**
29. What do you mean by molar and normal solution of NaCl? **(CMU:** Nov-24**)**

Isotope & Biotransport

1. What is isotopes **(CMU-May-19)** with example (May-17). (Ju-13, Nov-15) & classify it. (Ju-12,Ja-13) Give the medical use or use in medical research & science of isotopes? **(CMU-Nov-19)** (Ju-12,13 Ja-07, Nov-15, 16, May-17)
2. Define isotopes with example & give their uses in clinical medicine. **(CMU:** May-22, Nov-20) (Ja-11, Ju-14) Therapeutic use of isotope **(Nov-25) (CMU:** May-24**)**

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3. **Compare & contrast between**
   a) Active transport & facilitated diffusion.(Ju-07,08,09)
   b) Osmosis & diffusion. (Ju-09, Ja-15, 17)
   c) Absorption & adsorption. (Ju-09, Ja-06)
   d) Primary & secondary active transport. (Ju-11)
4. Classify biotransport mechanism across the cell membrane with example. **(CMU:** May-19). (May-17, 19). List the factors influencing the rate of diffusion (Ju-13, Ja-15, 17)
5. Draw and label a cell membrane. **(CMU:** May-19) (May-17)
6. **SN:**
   a) Isotopes **(CMU**: May-22, 20) (Ju-12 Ja-06,07, 14, May-15,16 Ju-15,16, Ja-16, Nov-16)
   b) Zwitter ion.(Ju-08)
   c) Epimer.(Ju-03,Ja-07)
   d) Radioactive isotopes(Ja-12)
   e) Secondary active transport.(Ja-12)
   f) Na^+ -K^+ ATPase pump.(Ju-07)
   g) Diffusion (May-16)
7. Define isomer. Discuss the isomer of monosaccharide. **(CMU:** Nov-21)

Bio Molecules

Carbohydrate

1. Define (Ju-12) & classify with example of carbohydrate? **(CMU:**Nov-23, 20, 19). (Ja-09,15, Ju-15, 16, Ja-17).
2. What are the reducing substance of blood. **(CMU:**Nov-23) State the principle of procedure of Benedicts test. **(CMU:**May-22) Non-carbohydrate reducing sugar **(CMU:**Nov-23)
3. Classify monosaccharide. (Ja-11,12,Ju-11)
   Mention the properties of carbohydrate with example. (Ja-07,Ju-12)
4. Name the isomers of glucose (CMU-Nov-20) with structural formula? (Ja-11, 12)
   Give the structure of -D. Glucose?
5. What is invert sugar? **(CMU: May-25)** (Ja-15)
6. **SN:** a) Mutarotation. **(CMU:**May-22, 21) (Ja-08, 11, 14, Ju-15, Ja-16, May-16)
   b) Anomeric carbon. (Ja-06)
   c) Isomerism. (Ju-09, 10,12 Ja-08, 06) d) Epimer. **(CMU:** May-20) (Ja-07)
   e) Benedict test.(Ju-10,12,13,14, 16, Nov-17)
   f) Principle of Benedict test **(CMU:** Nov-19) (May-19)
   g) Reducing sugar (Ja-03,04) h) Invert sugar (Ja-17)
   i) Anomer **(CMU:** May- 21)
7. What is reducing sugar? Write about the clinical importance of benedict's test. **(CMU:** Nov-24)
8. Define & classify play saccharides. **(CMU: May-25)**
9. Classify biomolecules with example. **(CMU: Nov-25)**

Protein

1. Define & classify protein **(CMU:**May-22) by its biological function. Give example from each. **(CMU:** Nov-24, 23,May-20) Classify amino acid? (Ja-09, Ju-10) What is denaturation of protein.**(CMU**: Nov-24, 23, 20)
2. Name the essential amino acids Give the properties of amino acid. (Ja-09,13 Ju-10, 11,12, Nov-16)
   Structure of amino acid in acidic, alkaline & neutral pH (Ju-12,13).
3. Give the properties of protein with biomedical importance. (Ja-08, Ju-13, Ja-17)
4. What are the structures of protein? **(May-25)** Draw a peptide bond. (Ja-12, May-18)
5. Mention the color reaction & important function of protein. (May-17)
6. Quaternary structure of protein. (Ju-14) What is Isoelectric p^H. **(CMU:** Nov-22)
7. Classify protein with example **(CMU:** Nov-20). List few biological important function of protein. (Ju-14)

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8. Draw & level a cell membrane. State the function of integral protein. (Ja-15, Ju-15, Nov-16)
9. Function of plasma protein with related applicable clinical importance (May-15, Nov-17)
10. Name some biologically important peptides. (Ja-17)
11. Name the bonds stabilizing the tertiary structure of protein **(CMU:** May-19) (May-17)
12. **SN:** a) Denaturation of protein.
    (CMU: May-22, 21, Nov-19) (Ju-12 Ja-11,14,15) with agent **(CMU:**Nov-23)
    b) Peptide bond. (J a-08, Ju-15, Nov-15, May-19)
    c) Heat coagulation test. (Ju-14, Ju-15, May-16, Nov-17) d) Isoelectric point/p^H.
    e) Zwitter ion. f) Amino acid (Ja-15) g) Microalbumin (May-16)
13. Show the structure of lipoprotein. What are the functions of different lipoprotein. **(CMU:**Nov-22)
14. Give the functional classification of protein with example. **(May-25)**
    Draw a peptide bond. **(CMU:**Nov-21)
15. Give the values f plasma profeins. List their important function. **(Nov-25)**

Lipid

1. Define & classify lipids with example? **(May-24, 23, 16,** Ju-10,11,Ja-13,14). Mention chemical/ Important properties of lipid. **(CMU:** May-22, 20) (Ja-12, 14,15, Ju-15, Nov-16, May-19)
2. Classify fatty acids? Mention the important properties of fatty acid?
3. Define & classify lipoproteins? **(CMU:** Nov-19) Give the structure of lipoprotein? (Ja-15, May-15,16 Nov-15) Give their biomedical & clinical importance & values? (Ja-11, 12,13 Ju-11,12)
4. What is prostaglandin? Name the prostaglandins. Show in a diagram the steps of synthesis of prostaglandins with their important function (Ju-12 Ja-09, 11, May-18)
5. Classify lipoprotein. Write the desired values of lipid profile. (Ju-16, Nov-17)
6. **Give the normal range of following-**
   a) Serum total cholesterol b) Serum TAG c) LDL d) HDL
   e) Serum albumin f) Serum A/G ratio. g) Total plasma glucose
7. Give the structures of TG & steroid nucleus (Ja-12). Name some derived lipids **(CMU:** May-22)
8. Lipid profile with importance. (Ju-06,12)
9. Draw & level the structure of chylomicron. What is the fate of chylomicron? (Ja-14, 19)
10. Write the biochemical importance of cholesterol. **(CMU:** May-23, 16, 15, Nov-19) (Ja-15, Nov-17)
11. Draw & level the structure of LDL. Classification of lipoprotein. (Ju-14)
12. How chylomicron is formed? (May-16)
13. Outline the functions of phospholipids **(CMU:** May-19) (May-19)
14. What is iodine number? **(CMU:** May-20)
15. **SN:** a) Specification number. (Ja-08, 07, 11, 12, Ju-15, 16)
    b) Iodine number. (Ju-06, 10, Ja-16, May-19)
    c) Acid number.
    d) Rancidity. **(CMU: May-25,** Nov-22, 21, 20) (Ja-07,09,13 Ju-13, Ja-16)
    e) Prostaglandin. (Ja-09, 11) f) Eicosanoids. (Ja-14)
    g) Poly unsaturated fatty acid (PUFA). **(CMU:** Nov-20) (Ju-14, May-18, 19, Nov-18)
    h) Phospholipid (Ja-15, Ju-15, Nov-15)
    i) Micelle (Nov-15) j) Cholesterol (Ju-16) k) LDL (Ja-17)
16. Name the essential fatty acid with their function. **(CMU:** Nov-24)
17. Show the structure of lipoprotein. What are the three common causes of secondary hypercholesterolemia. **(CMU:** May-24)
18. Classify fatty acids by hydrocarbon chain. **(CMU: May-25)**
19. Mention the importance of PUFA & trans fatty acid. **(CMU: May-25)**
20. How excess intake of carbohydrate deposit as fat. **(May-25)**

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BIOCHEMISTRY

Enzyme

1. Define & IUB classification of enzyme with example. **(Nov-25) (CMU:** Nov-23, 22, May-21) (Ja-09, 14, 15, Ju-10,12, Ju-15, May-16, 18, Nov-17)
2. Name the enzymes which help in diagnosis & prognosis of diseases (Ja-14, Ju-15, Nov-16).
3. List the properties of enzyme & **(Nov-25)** (Ja-06, 12, Ju-11, 16 Nov-15, Ja-16,17) major factors influencing enzyme activity. **(CMU**: Nov-20, 19) (Nov-17, 18),
4. Compare & contrast between competitive & noncompetitive inhibition of enzyme with diagram? **(CMU: May-25)** (May-23, Ja-09, 08, 07, 11, 12)
5. Differentiate between enzyme & co-enzyme? (Ja-06) Or, Co-enzyme & co-factor?
6. Give the clinical importance of enzymes (Ju-13)
7. Give few important examples of utilization of enzyme inhibition in drug designing. (Ju-14)
8. List the different types of enzyme inhibition (May-16). Explain with suitable example. **(CMU:** May-23, 22, 20)
9. Define isoenzyme with examples and their clinical applications? **(CMU:** May-21, Nov-19) (May-17)
10. **SN:** a) Allosteric effect. (Ju-10)
    b) Q_{10}. **(CMU**: Nov-19) (Ja-09, Ju-10,12, Ju-15)
    c) Km. **(CMU:** Nov-20) (Ja-07,15, Ju-10,12, Nov-15, 18, May-19)
    d) Co-factor. **(CMU:** Nov-20,22) (Ja-09)
    e) Iso-enzyme. **(CMU:** May-20) (Ja-08, 06, Ju-10, 11,12, May-17)
    f) Allosteric site (Ju-13) g) Co-enzyme (Ja-14,15)
    h) Enzyme inhibition **(CMU:May-25)**(Ju-14,May-23,15) i) Competitive inhibition (May-19)
11. What is coavalent odeification? **(CMU:** May-21) How substrate concentration influence enzyme activity **(CMU:** Nov-23,22**).**
12. Mention the factor influencing enzyme activity. What is Q-10? **(CMU:** 21)
13. Define enzyme. Discuss the factor influencing enzyme activity. Explain by graphical presentation. **(CMU:** Nov-24)
14. What is enxyme inhibitor name three enzyme inhibitors with their corresponding enzymes. **(CMU: May-25)**
15. Give the tools for nutritional assessment**. (CMU: May-25)**

GROUP-B

Food & nutrition

1. Define balanced diet? (May-23, 19), Give the criteria of balanced diet? **(CMU**-Nov-20) (Ja-08, Ju-14, Ju-15, Nov-15) **(CMU**-Nov-19)
2. What are the common nutritional disorders of Bangladesh? **(CMU:**Nov-23, Ja-06, 12, Ju-10)
3. What is PEM? Classify P'EM (Ja-14, May-15) Give the difference between kwashiorkor & marasmus? **(CMU:**May-23, 22, 21, Nov-23, 19) (Ju-10, 13) Give the features of Kwashiorkor (Ja-12, 14), Classify PEM (May-15,16, 18 Nov-16)
4. **Define BMI. How it helps in the diagnosis of-** a) Obesity (Ja-07, 11) What are the deficiency menifestation & common lab findings of it **(CMU:** Nov-23)
5. **Short note :**
    a) BMR (May-15, Ja-16)
    b) SDA **(CMU:** Nov-23, May-21) (Ja-07, Ju-11)
    c) Xerophthalmia **(CMU**-Nov-21, 20)
    d) BMI **(CMU**-Nov-22, 21, May-20) (Ju-07)
    e) Marasmus **(CMU**-Nov-21, May-20) (May-15,19, Nov-15, 16)
    f) Kwashiorkor **(CMU**-Nov-22) (May-15, Ju-16, Nov-17)

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g) PEM (Ja-09,13, Ja-17, May-18)
h) Obesity (CMU-Nov-20) (Nov-15, 17, 18, Ju-16)
i) Balance diet (CMU-May-19) (Ja-15) with criteria (CMU: Nov-23)
j) Trace elements. (CMU-Nov-20)
k) Malnutrition (Ja-08)
l) Fiber (Ja-16, 17) / Dietary Fiber (CMU-May-22, 21)

6. Name the dietary fiber. (CMU: Nov-23) What are the roles of dietary fiber in health & disease? (Ju-06, 13, May-16) (CMU-Nov-19) with clinical importance (CMU: May-23)
7. Define malnutrition & classify it (CMU-Nov-21) (Ju-13)
8. Define nutrition? Name some common causes of nutritional disorder with example. (Ja-15)
9. How will you prepare a balance diet for a sedentary female/male worker? (Age-22yrs, wt-55 kg or 60kg height-160 cm) (Nov-25) (CMU:May-21) (Ja-15, Ju-15, May-19)
10. Write down the welcome classification of PEM. Differentiate between major sub groups. (Ja-16, May-18)
11. What is the importance of fruits and vegetables addition in diet ? (Nov-16)
12. Write down the pathological causes of rise of BMR. Calculate the BMR of a 6 feet tall 75 Kg weight 20 years old (male) medical student. (Ja-17)
13. Define & classify BMI. (Ju-16)
14. Write the proximate principles of food. (CMU: Nov-23, 20, 16) (May-16, 19)
15. Define obesity. List the investigations required for diagnose PEM. (May-17), (CMU-Nov-19)
16. What are the different type of nutritional anemia? What is pernicious anemia.
17. Classify common nutritional disorder. (CMU-May-21)
18. What are the long term effect of undernutrition & overnutrition. (CMU-Nov-21)
19. Pt gave the history of per-rectal bleeding repeatedly during defecation physician advised him to take fiber containing diet.Explain how this kind of diet will reduce his symptoms?(CMU:Nov-23)
20. Calculate the energy requirement of 60 kg male medical student. (CMU: May-23)
21. State the energy requiring process in our body. Calculate the daily energy requirement of a 50kg female medical student. (CMU: May-25)

Vitamins and Minerals

1. **Write down the source, RDA and deficiency disorder / features, function of-**
   a) Vitamin-A (Nov-25) (CMU-Nov-19) (Ja-11,14,16,17 Ju-14, May-15, Nov-15, 18),
   b) Vitamin-D (CMU-May-22, 21, 20) (Ja-11, 12,13, 17 Nov-15, 17)
   c) Vitamin-C (CMU-Nov-21, 20, 19) (Ja-9, Ju-11, May-15, Ja-16, Nov-16, 17, 18)
   d) Vitamin- B_{12}.(Ja-13)
   e) Vitamin- B_1 (Ju-10)
   f) Folic Acid (Ja-12, Ju-13)
   g) Iron (Ja-14) (CMU-Nov-22)
   h) Xerophthalmia (Nov-16)
   i) Iodine (Nov-16)
2. Name the fat soluble vitamin with their active terms. (CMU:May-22)
3. What are the nutritional anemia? (CMU:Nov-22)
4. Write down the function of- "Vitamin-A (Ju-09, Ja-14, May-15, Ja-16) Vitamin-D (Ja-09, Ju-15) Vitamin-C. (May-15, Ja-16)
5. Classify water soluble vitamins. Mention their biological importance (Ja-07)
6. Define and classify minerals with ex. (Ja-17 Ju-09, 13)
7. Show in a diagram the metabolism & absorption of iron? (Ju-09, 11, 17, May-16)
8. Show in a diagram the regulation of blood Ca^{++} level with normal level of Ca^{++}. (Ja-15, Ju-10,14) Function of calcium.

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9. Give the source, deficiency syndrome, RDA, function & metabolism of iodine. (Ju-13, 16)
10. Give sources, function & deficiency features (May-19), of iron, zinc & Vitamin- C (May-19) (Ju-10, May-16) Vitamin-C & Iodine **(CMU:** Nov-23)
11. Name the water soluble vitamin with their active form. **(CMU:** Nov-23) Mention the sources, RDA, function, deficiency features of vitamin B_{12} & folic acid **(CMU:** Nov-23, May-22) (Ju-13, Ja-17)
12. Define & classify vitamin. **(CMU:**Nov-23, 21) (Ju-14, Nov-16)
13. How vit-A associated with health and disease. (May-15)
14. What is trace element? **(CMU:**May-22) Write down the source and function of iodine in the body. (Ja-16) importance of iodine in nutrition (Nov-17)
15. Steps of synthesis of vit-D **(Nov-25)** (Ju-15)
16. Write down the role of Zinc, Iodine and iron in heath & diseases **(CMU:**May-23, 22, Nov-19) (May-16)
17. What are the extra ocular features of vitamin A deficiency (Ja-17)
18. What is folate trap mechanism? **(CMU:** Nov-23, Ja-17)
19. Name the factors influence iron absorption (Ja-17)
20. Define & classify trace elements **(CMU:** May-23, 16, Ju-16, Nov-17)
21. **SN:** a) Rickets **(CMU:**May-22) (Ju-06)
    b) Trace elements (Ju-08,11, Ja-11,17, Nov-15, May-19) **(CMU-**May-19)
    c) RDA (Ja-08, Ju-08) d) Beriberi
    e) Micronutrients (Ju-07) Micro-mineral.
    f) Anti oxidant vitamin **(Nov-25)** (Ja-14, May-19)
    g) Scurvy (Nov-15, Ja-15, May-17) h) Pellagra (Ju-16)
    i) 7-dehydrocholesterol **(CMU-**May-19)
    j) Vitamin-K
    k) Vit-A (May-19) l) Hypokalemia **(CMU-**May-21)
22. Name the hemopoitic and energy related vitamins. How Vit-C is associated with post translational modification. **(CMU: May-25)**
23. Write about co-enzyme forms of different vitamins. **(Nov-25)**

DIGESTIVE SYSTEM

Local hormone of GIT

1. List the local hormones of GIT. Give the function of any two of them. (Ja-09, 11, 12, Ju-10)
2. **Discuss source, cause of secretion & function of-**
   a) Gastrin (Ja-09, 08, 07, Ju-16)
   b) CCK-PZ (Ja-08,15)
   c) Local hormones(Ju-12 Ja-13)
   d) Renin (Ja-15)
3. **S/N:** GI hormon. **(CMU-**Nov-20)

Digestive Juice

1. Name the digestive juices? **(CMU:**Nov-22) (Ju-13, Ju-16) with pH? **(May-25)**
2. **Give the composition & function of following:**
   a) Gastric juice. **(CMU:** Nov-22, May-20)) (Ja-07,13, Ju-10, May-15, 19, Nov-15, 18)
   b) Pancreatic juice (Ja-05, 10, 14, Ju-07) **(CMU-**May-19)
   c) Bile (Ja-09, Ju-09)
   d) Succus entericus (Ja-08)

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